

Augmented Haptic VR Experience Combining Two Weight-Shifting Versatile Controllers

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ABSTRACT

We designed a VR controller to integrate experimental haptic technology into a practical controller. The device consists of two independent controllers, each with a weight-shifting module that can provide vibration, impact, and shape perception yet is sufficiently compact to be handled as a conventional commodity controller. Combining two controllers allows the device to be held differently for various applications.

CCS CONCEPTS

• Human-centered computing → Haptic devices; • Computing methodologies → Virtual reality; Perception.

KEYWORDS

Haptics, Controller, Weight shifting, Vibration

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1 INTRODUCTION

In the early 2020s, many head-mounted displays entered the market, and with them came many VR controllers. The primary function of these commercial handheld controllers is to track hand position and provide a physical input interface. Haptic feedback is sometimes implemented to enhance immersion and usability, but in many cases, it is just a small vibrator, so the range of expression is limited.

Some solenoid-based actuators on the market can extend expression for impact presentation. However, noisy sound, high price, and unwieldy size are issues for installation in a versatile commodity controller.

Furthermore, VR experiences sometimes involve holding objects larger than the palm, such as a sword or gun, which often leads to

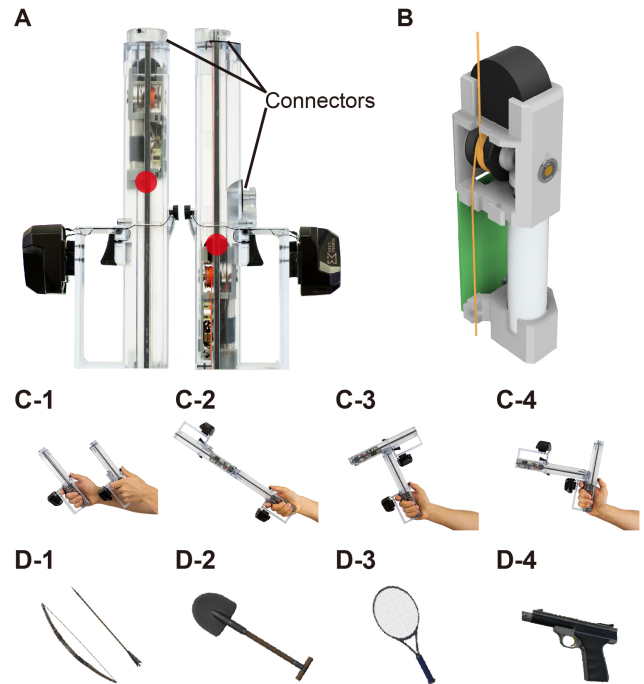


Figure 1: (A) The prototype is two weight-shifting controllers, with a weight-shifting module inside and connectors outside. Red points represent their COGs. (B) The weight-shifting module consists of a motor, board, gears, reel, and voice coil motor. (C) The device can be held in four different ways, (D) enabling a wide range of applications.

a lack of immersion due to a mismatch in sensation between the controller and the virtual object the user is holding.

In solving this problem, Shifty [Zenner and Krüger 2017] demonstrated that a single cylindrical controller could render different lengths and weights of virtual objects by moving the center of gravity (COG) up and down. Transcalibur [Shigeyama et al. 2019] extended this concept to two dimensions. Drag: On [Zenner and Krüger 2019], ShapeSense [Liu et al. 2019], and Unident [Shimizu et al. 2021] realized the sensation of resistance or impact by focusing on the reaction force when an object is wielded. However, because these shape display devices are specialized for each device, challenges remain concerning typical VR experiences.

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Therefore, we listed the requirements for a versatile controller that can be used for typical VR experiences as follows:

- One in each hand
- Simple configuration, neither too large nor too heavy
- Maintain the COG within the palm
- Functions of a conventional VR controller, such as input buttons and haptic vibration

In this study, we developed a versatile haptic controller that satisfies these requirements and can present vibration, impact, and shape. We demonstrated that the previously proposed haptic experiences could be integrated into a practical controller.

2 PROTOTYPE

Controller Specifications. The prototype is a pair of VR controllers, each with a weight-shifting module that changes the device's COG (Fig1: A). They are 250 mm long to be safe enough to wield and have enough change in their physical property. They are 29×34 mm in diameter (equivalent to the grip of a tennis racket) and 220 g in weight, including a tracker. They have an index finger trigger and thumb button as input interfaces.

Weight-Shifting Module. The weight-shifting module consists of a motor, control board, miter gear, reel, and voice coil motor (VCM), with a total length of 80 mm and a weight of 70 g (Fig1: B). Increasing the ratio of the module's weight to the total device weight is necessary to increase the efficiency of changing the COG position in the limited weight-shifting distance. In this device, we achieved it by combining all the components necessary for the drive in the weight-shifting module. The module can be lowered to the bottom end of the device, and the COG of the entire device at that time is in the palm, avoiding generating unnecessary moments during regular use without shape presentation.

Actuation. The reel winds fishing lines stretched inside the device, allowing the weight-shifting module to move up and down at a maximum speed of 120 cm/s. The COG can be adjusted within a range of 84 to 133 mm from the bottom of the device. This fast motion facilitates quick weight-shifting and allows the user to experience vibration by moving the weight-shifting module up and down at high speeds. Combining low-frequency vibration with vertical movement of the weight-shifting module and high-frequency vibration with the VCM makes it possible to create rich tactile expressions ranging from a strong impact to elaborate vibration (Fig2).

Combination. Connectors are mounted on the top and side of the controller. Two devices can be easily connected in two configurations (I- or T-shaped) while wearing the HMD (Fig1: C). Thus, it is possible to render a larger or more complex virtual object than with a single device (Fig1: D).

3 APPLICATION

At SIGGRAPH 2023 Emerging Technology, we will demonstrate the following VR applications.

Split Mode. When separated, the user holds a controller in each hand (Fig1: C-1). By keeping the weight-shifting module at the bottom, the user has the same experience as a conventional commodity

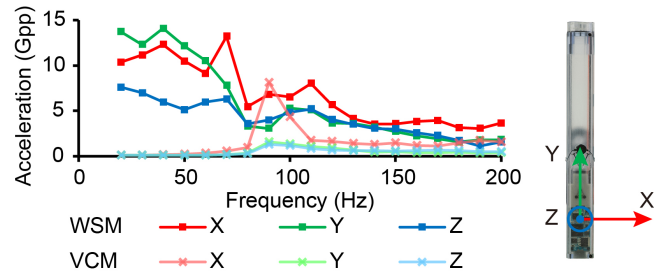


Figure 2: Acceleration-frequency characteristics of vibration by weight-shifting module (WSM) and VCM. IMU was attached to the grip of the device.

VR controller and feels the mass and impact of a virtual object by moving the weight-shifting module up and down or vibrating the vibrotactile actuator. Because both controllers move independently, it is possible to express, for example, holding a bow in the left hand and an arrow in the right.

I-Shape Mode. By connecting the top ends of the controllers vertically, the controller becomes I-shaped (Fig1: C-2), making it possible to render long virtual objects. Compared with a single device, the maximum distance from the COG to the device's bottom edge is approximately two times longer, 269 mm, and the weight is two times heavier. Hence, the total moment is four times greater. Furthermore, by shifting the COG, it is possible to express the weight, for example, of how much soil is scooped up with a shovel.

T-Shape Mode. By connecting the top and side connectors, the controller becomes T-shaped (Fig1: C-3) to permit displaying a virtual object with a heavy tip, like a hammer. Furthermore, the impact of a sword strike or a tennis racket can be displayed by quickly moving the weight-shifting module left or right with one pulse.

Rotated T-Shape Mode. The controller becomes gun-shaped when connected in a T-shape and rotated (Fig1: C-4). The size of the gun can be expressed by moving the COG horizontally, and the feeling of shooting can be presented by vibrating it at high speed.

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